

Active Directory RBAC & Delegated Administration Lab

Proxmox VE • Windows Server • Active Directory • RSAT • Access-Control Validation

Portfolio Project | mattlab.local

1. Project Overview

This project built and validated a small enterprise-style Active Directory environment inside a Proxmox homelab. The work began with a Windows Server virtual machine that became the DC01 domain controller for mattlab.local, continued through OU, user, group, password-policy, and delegated-permission configuration, and concluded with a separate Windows client VM used to test the controls from an endpoint rather than directly on the domain controller.

The central goal was to demonstrate role-based access control (RBAC) and least privilege. Administrative users were given scoped authority through security-group membership and Delegation of Control, while standard users were intentionally denied the same administrative actions.

2. Objectives

- Deploy the Active Directory lab as virtual machines on Proxmox VE.
- Configure Windows Server as a domain controller and DNS server for mattlab.local.
- Create departmental OUs, security groups, and representative user accounts.
- Apply domain password requirements and verify they are enforced.
- Delegate specific administrative tasks to role-based admin groups instead of granting unrestricted domain-wide rights.
- Build a separate Windows client, connect it to the isolated lab network, join it to the domain, and install RSAT.
- Perform positive and negative access-control tests from the client to prove the delegation works as intended.

3. Lab Architecture

Component	Platform	Role	Key configuration
Proxmox host	HP EliteDesk 800 G4 Micro	Hypervisor	Hosts the lab VMs.
DC01	Windows Server	Domain Controller / DNS	mattlab.local; lab IP 10.10.10.10.
Windows-Client	Windows Server used as a client/test workstation	Domain-joined management endpoint	4 GB RAM, 32 GB disk, VirtIO NIC, vmbr1, RSAT.
Lab network	Proxmox vmbr1	Isolated VM network	10.10.10.0/24; client and DC communicate directly.

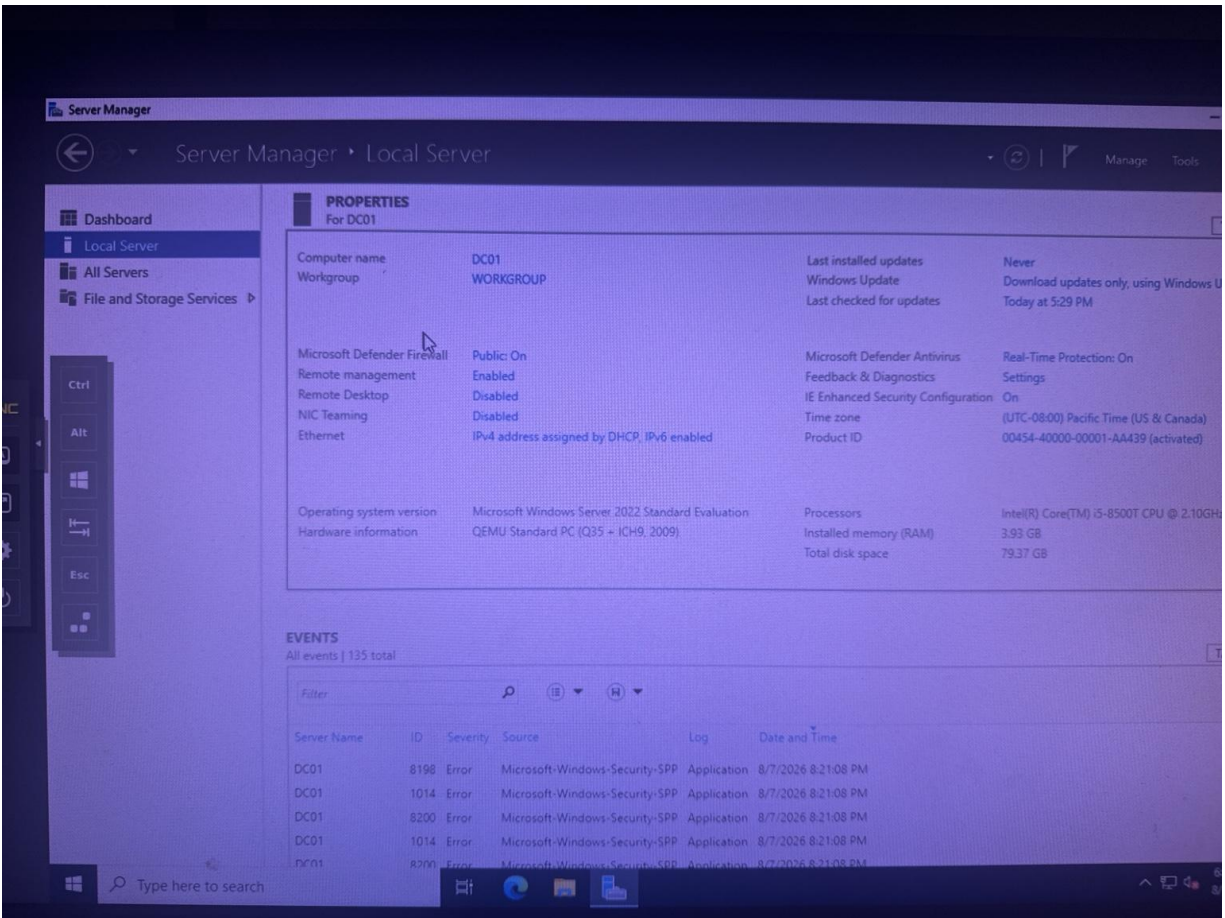


Figure 1. Proxmox environment used to host the Windows Server lab virtual machines.

4. Domain and Active Directory Configuration

The Windows Server VM was configured as the domain controller for mattlab.local. Active Directory Domain Services and DNS were used to provide centralized identity, directory, and name-resolution services for the lab. The directory was organized into departmental OUs including IT, Finances, and HR.

Representative users were created and assigned to role-based security groups. The six primary lab users were reviewed before validation: LeBron, Steph, and Tim were assigned administrative roles, while Nash, Chuck, and Vince were standard users. This provided distinct identities for testing authorization boundaries.

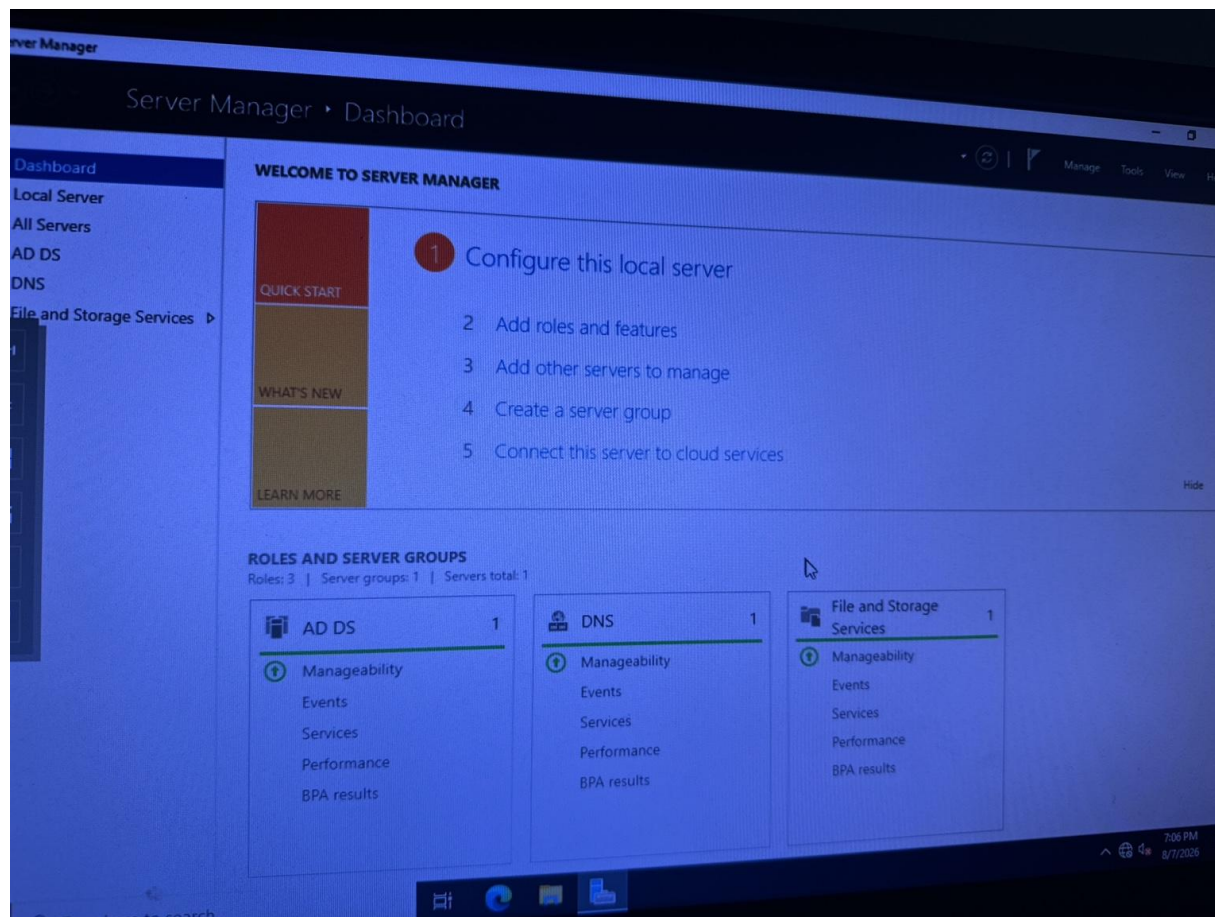


Figure 2. Windows Server / Active Directory services operating on the domain controller.

5. Password Policy Enforcement

The domain password policy was configured with stronger requirements, including a 12-character minimum length, 90-day maximum password age, 1-day minimum password age, 20-password history, and reversible encryption disabled. During testing, Active Directory rejected a password for a disposable test account because it did not satisfy the configured policy. An existing administrative account also required a password change, providing practical evidence that the policy was being enforced rather than merely configured.

6. RBAC, OU Permissions, and Delegation of Control

Security groups such as IT_Admins / IT_Users and Finance_Admins were used to represent job roles. OU permissions were reviewed and corrected before final testing. Delegation of Control was then applied to the administrative groups at the appropriate OU scope.

For IT_Admins, delegated tasks included creating, deleting, and managing user accounts; resetting user passwords and forcing password changes; and reading user information. This allowed a delegated administrator to manage users in the IT OU without making that account a Domain Admin.

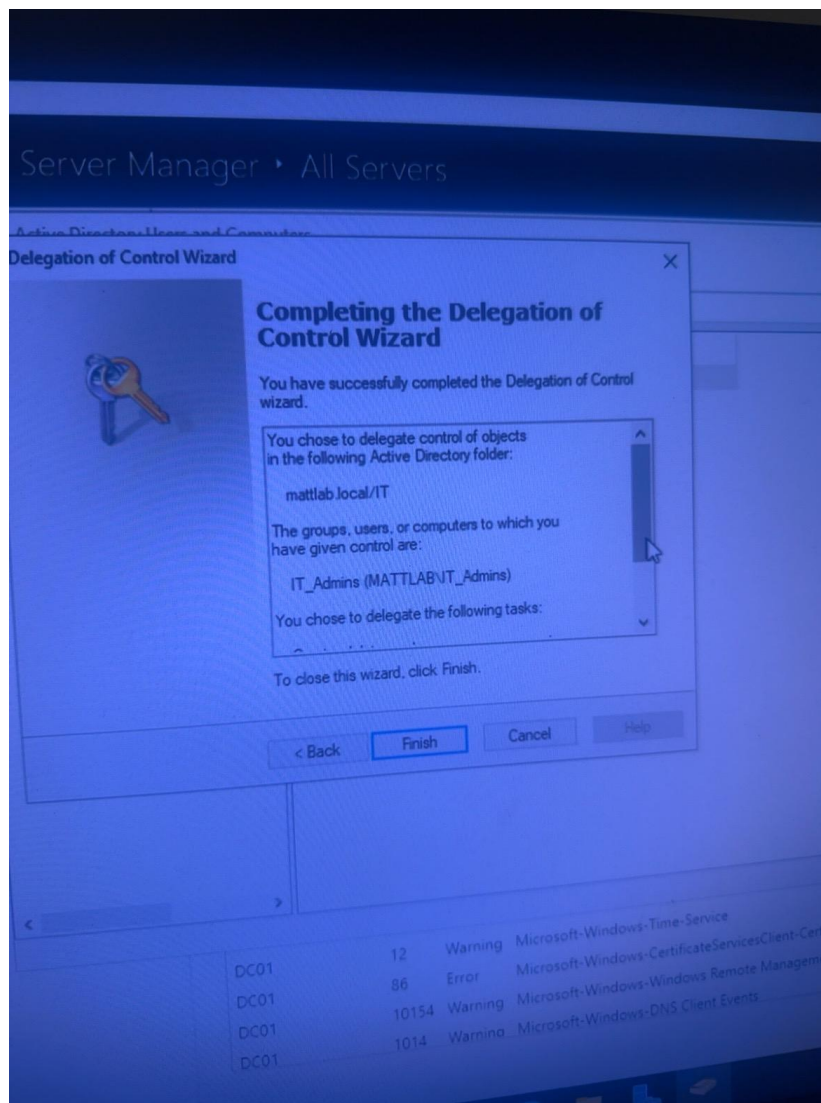


Figure 3. Delegation of Control completed for the IT_Admis security group, including user-account management tasks.

A similar scoped delegation was configured for the Finance administrative role. The project also included reviewing advanced security permissions on the OUs so that the intended group-based permissions were present before validation.

7. Building the Separate Windows Client VM

To test the environment realistically, a second Windows VM was created in Proxmox rather than performing all validation locally on DC01. The VM was provisioned with 4 GB of memory, a 32 GB virtual disk, UEFI/OVMF, a VirtIO SCSI disk, a VirtIO network adapter, and the vmbr1 bridge. QEMU guest-agent support was enabled so the VM could integrate cleanly with Proxmox.

The Windows installation initially lacked a functional network adapter because the VirtIO driver was not yet available inside the guest. The VirtIO driver ISO was mounted and the guest tools/network driver were installed. This changed the client from having no usable lab-network configuration to exposing the Red Hat VirtIO Ethernet Adapter.

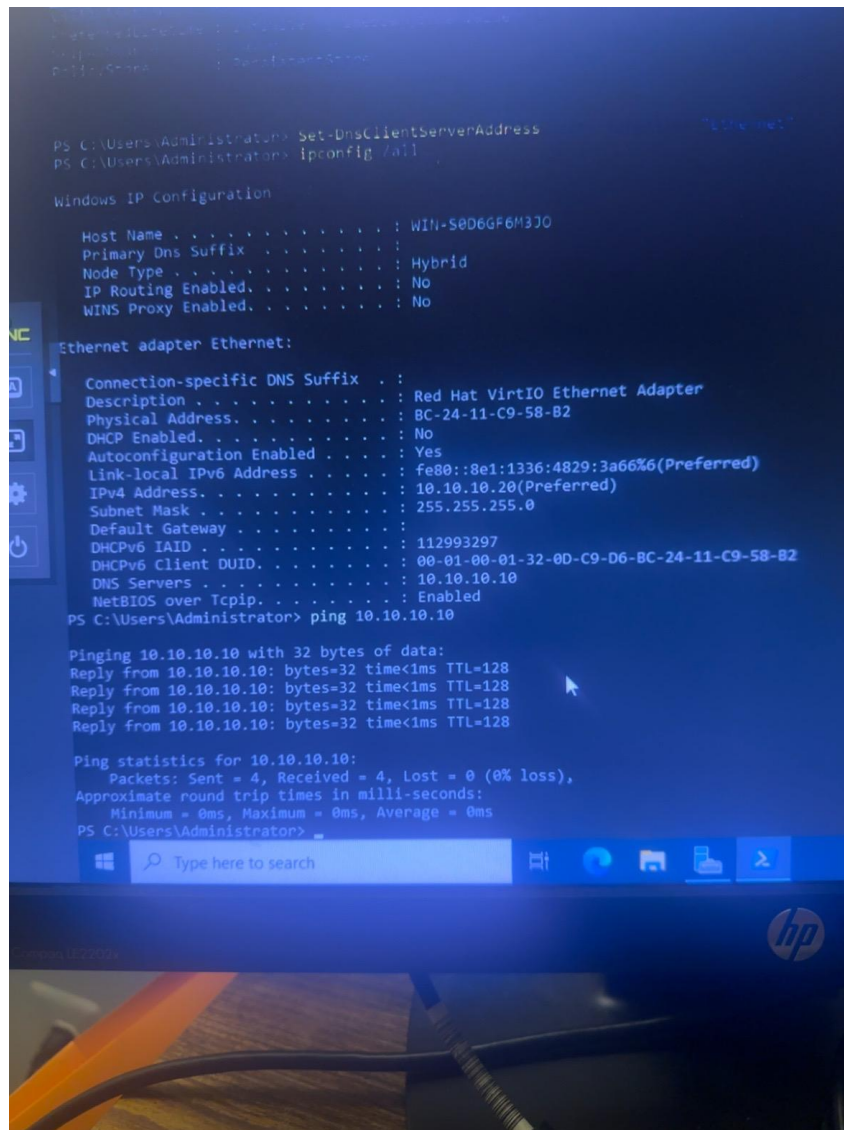


Figure 4. Windows client after the VirtIO network adapter was installed and became available to the operating system.

8. Network Troubleshooting and Static Configuration

The client initially self-assigned an APIPA address in the 169.254.0.0/16 range. At the same time, DC01 was configured on 10.10.10.10/24. Because the two machines were not on the same IPv4 subnet, ping tests failed. This was a useful troubleshooting checkpoint: the virtual NICs and vmbr1 connection existed, but Layer 3 addressing was inconsistent.

The client was then manually configured as 10.10.10.20/24 and its DNS server was set to 10.10.10.10 (DC01). After the change, the client successfully pinged 10.10.10.10 with 0% packet loss. This confirmed basic IP connectivity before attempting the domain join.

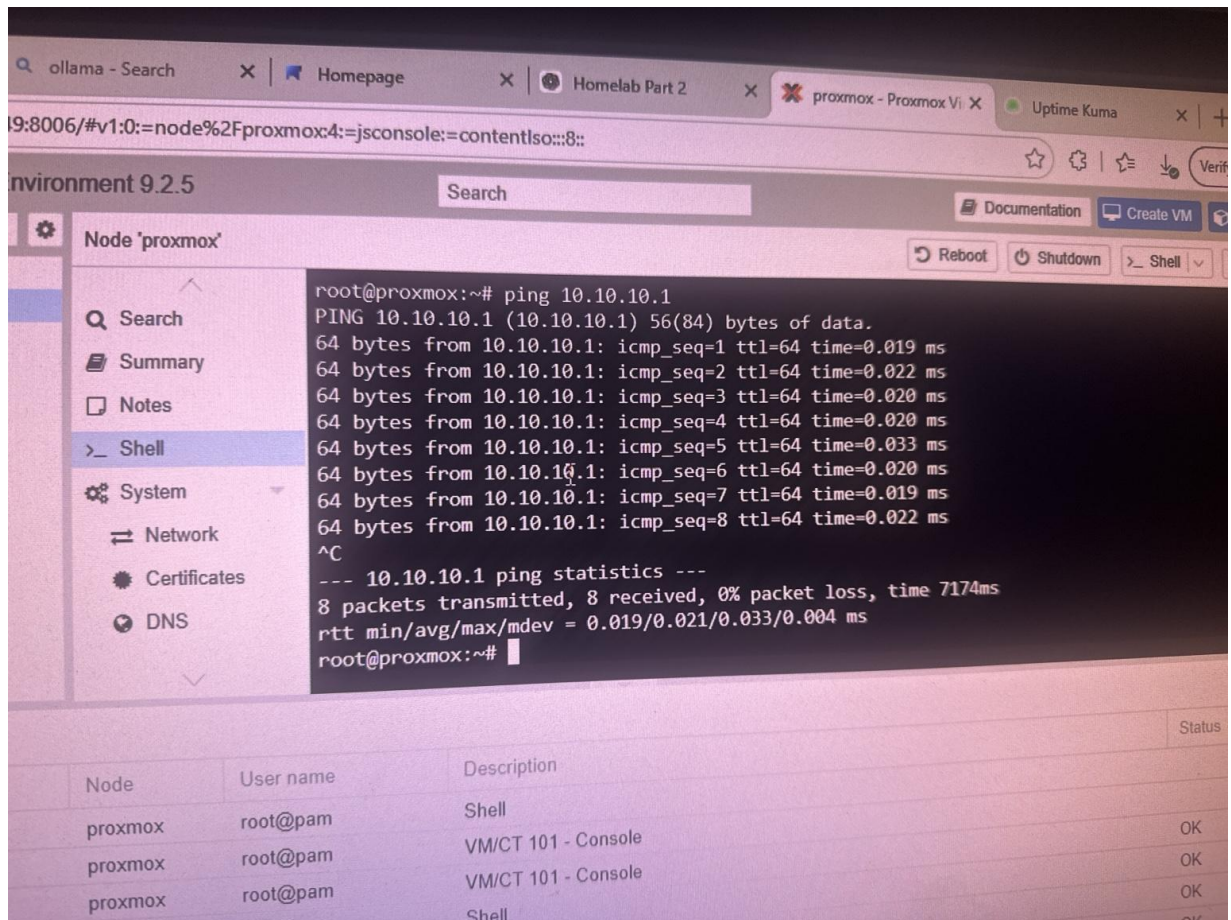


Figure 5. Successful client-to-DC01 connectivity test after correcting the client IPv4 configuration.

9. Joining the Client to mattlab.local

With IP connectivity and DNS pointing to DC01, the Windows client was renamed Windows-Client and joined to mattlab.local. The domain join completed successfully and the VM was restarted. Domain accounts could then authenticate to the client; LeBron was used as the delegated administrator for the positive validation test.

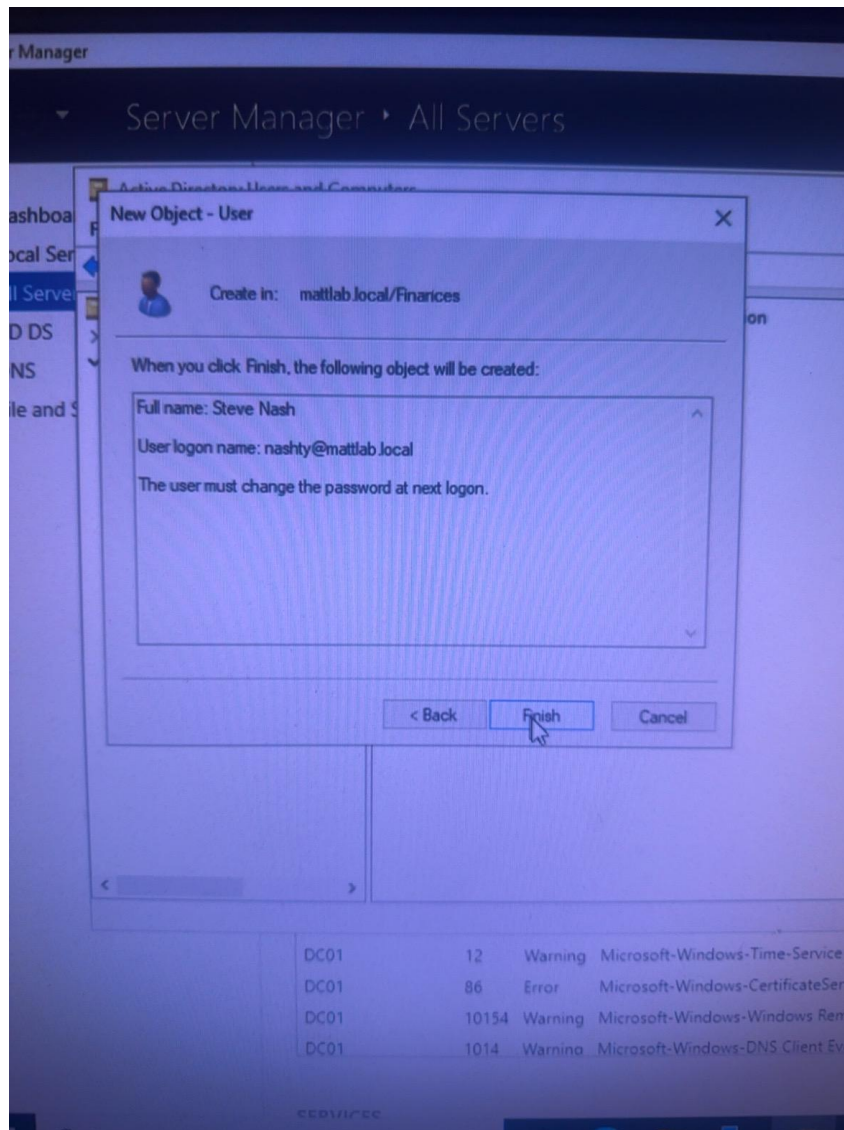


Figure 6. Windows-Client successfully joined to the mattlab.local domain.

10. Installing RSAT for Remote Administration

The next requirement was Active Directory Users and Computers on the client. Only the management tools were needed; the client was not promoted to another domain controller. AD DS and AD LDS Tools under Remote Server Administration Tools (RSAT) were selected so the existing domain could be managed remotely.

An installation attempt while signed in with the delegated LeBron domain account failed with “You do not have adequate user rights to make changes to the target computer.” This highlighted an important distinction: delegated Active Directory rights do not automatically provide local administrator rights on a workstation. The RSAT feature was therefore installed using the client’s local Administrator account, after which the delegated domain user could use ADUC.

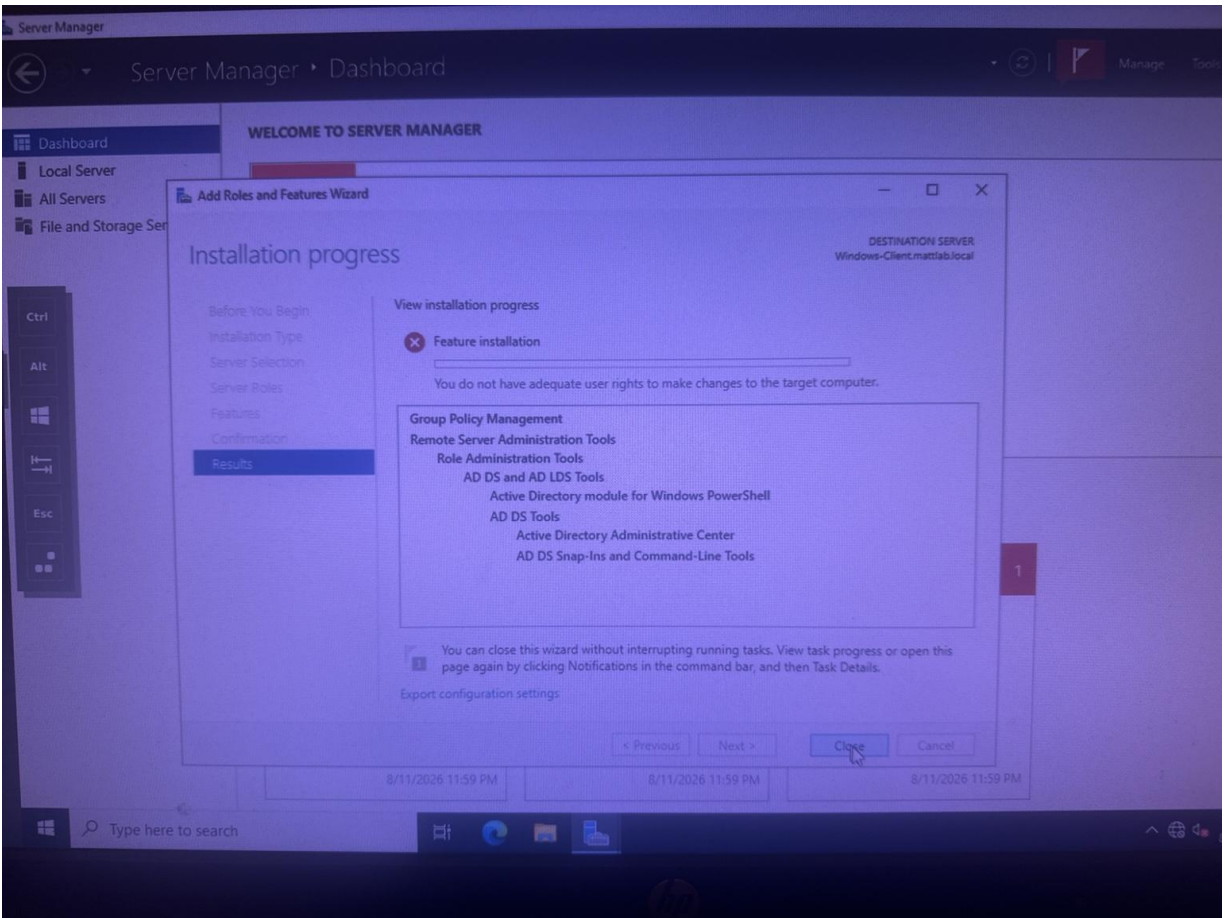


Figure 7. RSAT installation initially blocked because the delegated domain account did not have local administrative rights on Windows-Client.

11. RBAC Validation

11.1 Positive Test - Delegated IT Administrator

A disposable Test User 01 account was created in the IT OU. While signed into Windows-Client as LeBron, a member of IT_Admns, Active Directory Users and Computers was opened through RSAT. LeBron was able to delete Test User 01 successfully. This verified that the IT_Admns delegation worked remotely from a domain-joined endpoint.

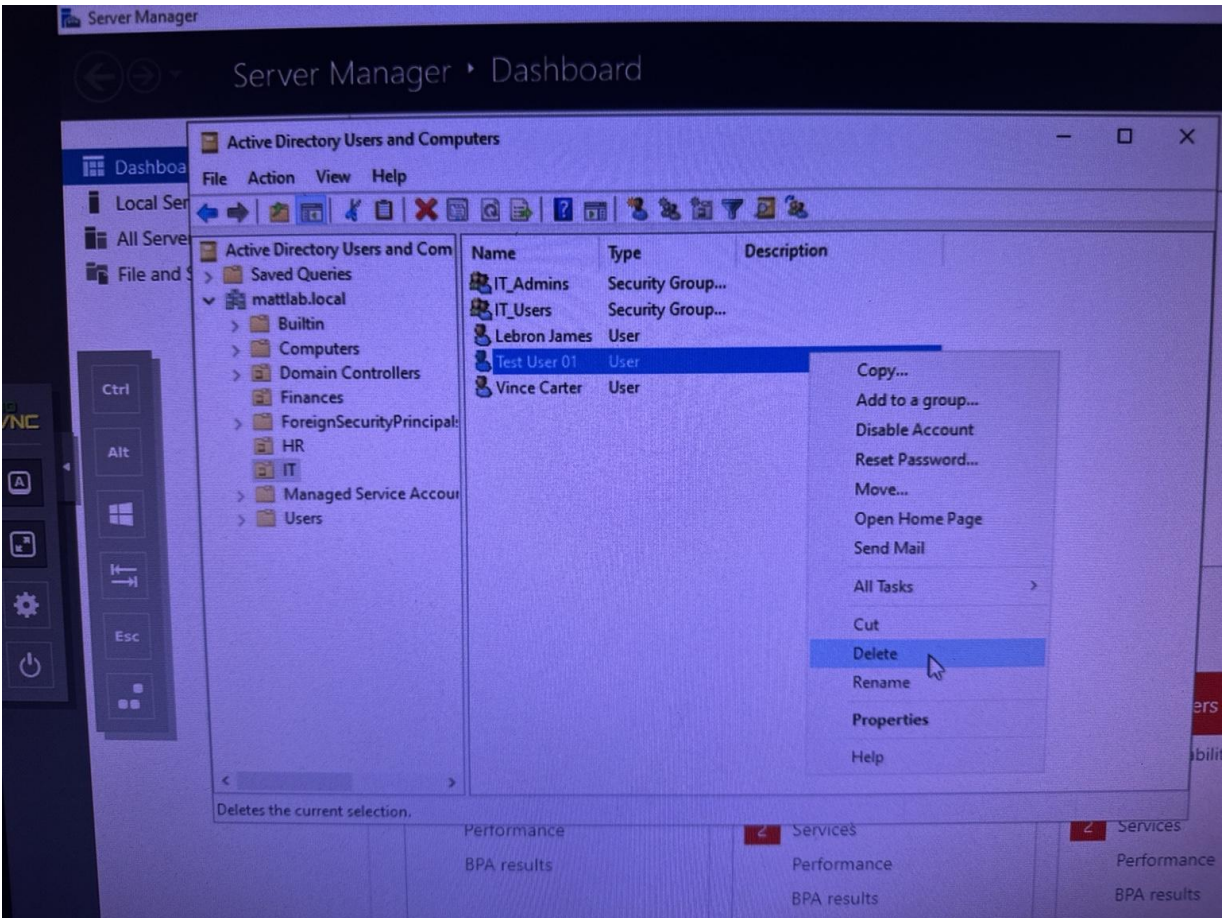


Figure 8. Positive authorization test: LeBron, through IT_Admins delegation, is able to initiate deletion of Test User 01 in the IT OU.

11.2 Negative Test - Standard User

A second disposable account, Test User 02, was created. The client was then signed into Vince, a standard IT_Users account. Vince attempted the same delete operation. Active Directory returned an insufficient-privileges error and Test User 02 remained in the directory. This demonstrated that the standard-user role did not inherit the delegated administrative permissions.

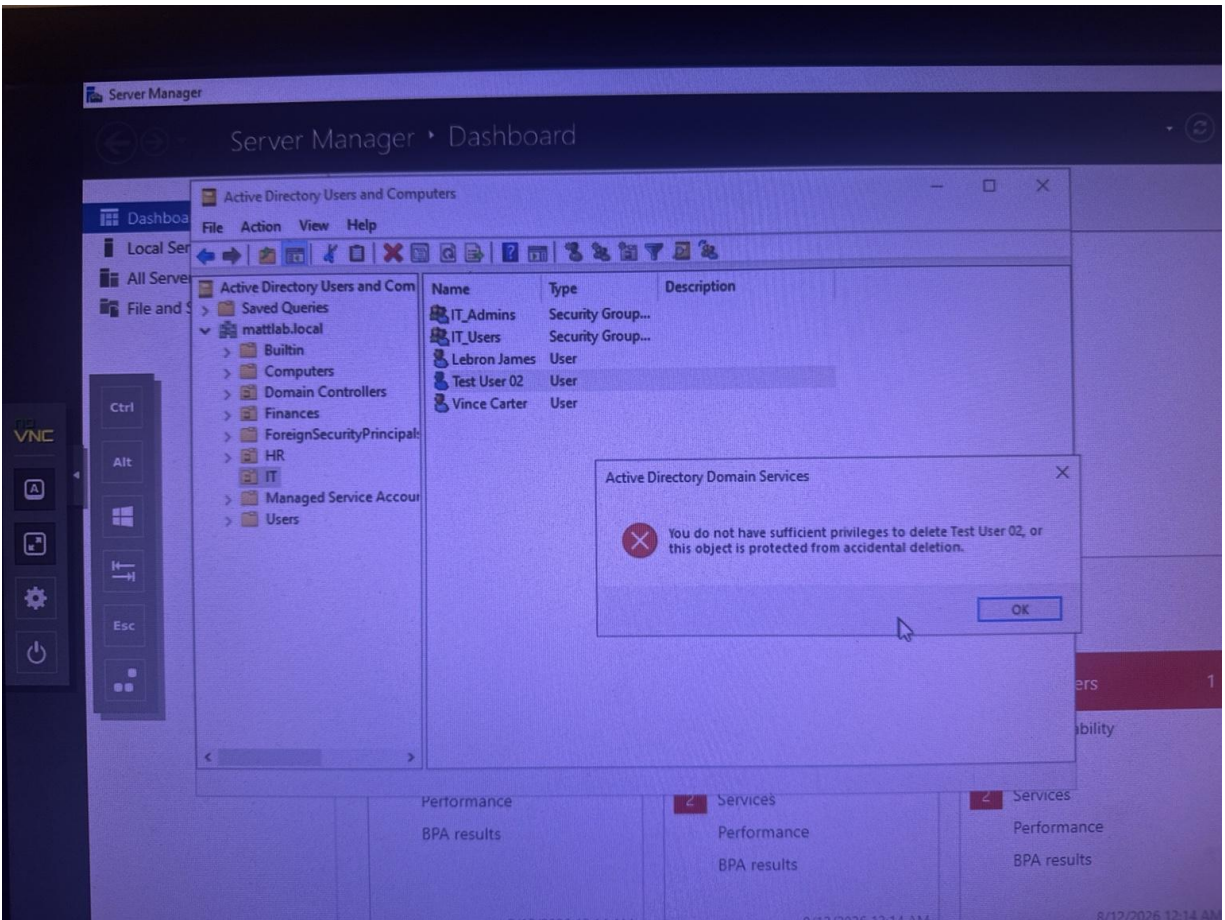


Figure 9. Negative authorization test: Vince is denied permission to delete Test User 02.

12. Validation Results

Identity	Role	Test	Result
LeBron James	IT_Admins	Delete disposable user in IT OU	PASS - authorized action succeeded.
Vince Carter	IT_Users	Delete disposable user in IT OU	PASS - unauthorized action was denied.

13. Troubleshooting Summary

Several issues occurred during the build, but only the items that materially affected the final implementation are summarized here. Some exploratory troubleshooting and repeated checks were intentionally omitted from this portfolio version.

- **VirtIO networking:** the new Windows VM initially had no usable virtual network adapter until the VirtIO guest/network drivers were installed.
- **Subnet mismatch / APIPA:** the client received a 169.254.x.x address while DC01 used 10.10.10.10/24, so the systems could not communicate. Assigning the client 10.10.10.20/24 resolved the addressing mismatch.
- **DNS dependency:** the client was configured to use 10.10.10.10 as DNS so it could resolve mattlab.local and locate the domain controller.

- Password policy: a weak test password was rejected, confirming that the configured domain password policy was active.
- Local versus delegated privilege: LeBron could administer delegated AD objects but could not install RSAT locally. Local Administrator credentials were used for the feature installation without broadening LeBron's workstation privileges.

14. Skills Demonstrated

Proxmox VE • Windows Server administration • Active Directory Domain Services • DNS • Organizational Units • Security groups • RBAC • Delegation of Control • Least privilege • Password policy • Windows domain joining • RSAT • Active Directory Users and Computers • VirtIO drivers • Static IPv4 configuration • VM networking • Connectivity troubleshooting • Positive/negative access-control testing

15. Project Outcome

The project produced a functioning multi-VM Active Directory environment and, more importantly, validated the security design through observable behavior. A delegated administrator could perform the intended account-management action, while a standard user attempting the identical action was denied. The separate Windows client, domain join, RSAT deployment, and network troubleshooting made the validation representative of a real administrative workflow instead of a configuration-only demonstration on the domain controller.